



ZSROI

Generic Integration

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Introduction

This document is intended for Zone Soft partners who want to integrate their services using ZSROI to send orders directly to the POS.

This document aims to define the communication, authentication, menu structure and order structure in Zone Soft integration.

It aims to define endpoints that integrators should use, the structure and types of response expected from each, as well as the endpoints that integrators should implement, the structure and their responses.

Communication and Authentication

First of all you need to register a user in <https://developer.zonesoft.org> and create a 'New Integration' (please see "Platform Manual" document).

Communication with the integrator will be via endpoints defined when creating a new integration.

Below is an example of the structure of a generic integration

```
{
  "app_key": "YOUR-KEY",
  "app_secret": "YOUR-APP-SECRET",
  "app_name": "YOUR-APP-NAME",
  "endpoints": {
    "endpoint_login": "https://example.com/auth/login",
    "endpoint_menu": "https://example.com/sync/menu",
    "endpoint_order_status": "https://example.com/order/status",
    "endpoint_pos_status": "https://example.com/pos/status/"
  }
}
```

endpoint_login - It's for API-ZSROI authenticate in Integrator's app with integrator APP- KEY and APP-SECRET (will be generated when you create a 'New Integration' in <https://developer.zonesoft.org/login>) that should return a token.

The token will be sent in headers to the others endpoints (see section Login).

endpoint_menu - This endpoint is for sending the menu (see section Menus).

endpoint_order_status - This endpoint is for send the Orders Status (see section Endpoint_order_status)

endpoint_pos_satus - This endpoint is for send information about POS status (see section Endpoint_pos_status)

ZoneSoft's integrator is responsible for creating the necessary endpoints to enable communication between their application and ZoneSoft. While the provided endpoints serve as an example, the integrator must implement their specific endpoints.

Where endpoints need to be configured

After creating a 'New Integration' in <https://developer.zonesoft.org> you need to create a new client and set all endpoints to finalize the integration with API-ZSROI.

New Client

The Zone Soft's client must have the proper license active on the STORE ID for the API you're connecting to.

Client Serial

Client Serial

Endpoint login

Endpoint login

Endpoint order status

Endpoint order status

Store ID

Store ID

Permissions

Select a permission

Endpoint menu

Endpoint menu

Endpoint pos status

Endpoint pos status

Url Logo 

Url Logo

The image must be png and with maximum size 800x400

☐ I agree to the [terms](#)

CancelSave

If you don't have a Client Serial you should to contact our commercial department (comercial@zonesoft.org)

Headers to send in request to API-ZSROI

You need to add this information in the request to communicate with API-ZSROI when: Send an order - (see section Orders)
Get Tables - (see section Get Tables)
Reservation table - (see section Reservation table) Please see the example files and the postman collection.

Authorization header you must send your APP-KEY found on the portal:
<https://developer.zonesoft.org/login>

X-Integration-Signature header you must send the request signature with your APP- SECRET that can also be found on the portal.

Signature:

```
hash_hmac('SHA256', $payload, $app_secret);
```

payload - request body

app_secret - your APP-SECRET found on the portal: <https://developer.zonesoft.org/login>

Login

For ZSROI to synchronise the menu with the integrated App or access any other functionality provided by the integrated App, an authentication token must be obtained.

ZSROI will obtain that token through an authentication request to the endpoint defined for the App when it was registered on Zone Soft's integrator platform.

Example endpoint: `https://example.com/auth/login`

The token obtained will be sent in the header of each request to access the integrated App. The App must validate the token received.

Endpoint_login (defined in the integrator platform for the App)

```
curl--location--request POST 'https://example.com/auth/login \
--data-raw '{
    "app_store_username": "YOUR_APP_KEY",
    "app_store_secret": "YOUR_APP_SECRET"
}'
```

Example of expected response

```
{
  "body": {
    "access_token": "TOKEN",
    "expires_in": 2592000
  },
  "header": {
    "statusCode": 200,
    "statusMessage": "OK",
    "status": "HTTP/1.1 200 OK"
  }
}
```


Menus

Menus will have the functionality to set menu products in the integrated App, as well as shop hours and status. Create, update menus and set the shop as online or offline in the integrated App.

Menus will be built by the client/restaurant in ZSBMS, being necessary to activate the products to be available in the integration or using digital menus, assigning a specific menu to the integration.

Json structure of the Menu

The top-level menu object has these keys: families, products, attribute_groups, attributes, packagings. subfamilies are nested inside each family (they are not a top-level array).

- **families (array)**

A family is the main category of a menu. It is mandatory to have at least one family on the menu. Examples of families: Drinks, Burgers, Salads, Desserts, Pasta.

In Zone Soft represents the table of *familias*.

- **subfamilies (array)**

A subfamily is a subdivision within families. They are not compulsory. Examples of sub-families: Soft drinks (family: Beverages), Carbonara (family: Pasta).

In Zone Soft represents the table of *subfamilias*.

- **products (array)**

A product is the main component of a menu. Each product belongs to a family and/or family + sub-family.

Zone Soft represents the table of *produtos*.

- **attributes (array)**

A product can be personalized in different ways. For example, adding a drink or adding an extra ingredient. Each of these customisations is considered as an attribute. It is not mandatory for a product to have attributes. Each attribute can have a different price impact. Attributes from the same menu are grouped into attribute groups.

In Zone Soft represents the table of *niveismenu*.

- attribute_groups (array)

Attribute groups allow you to group similar customisations into generic groups. Each attribute group can have a minimum number and maximum number of choices allowed to the client.

Examples of attribute groups:

- Side dish: rice and/or potatoes
- Drinks: water, coca-cola

In Zone Soft represents the table of *niveismenuext* and *complementares*.

- packagings (array)

Packagings are used to identify products subject to the Deposit Return System (DRS) in Portugal. This field determines whether a deposit applies to a product and enables its handling during sale and return processes.

Structure of the array families

Field name	Type	Description
name	string	Family name
subfamilies	array	List of sub-families

Structure of the array subfamilies

Field name	Type	Description
name	string	Name of the subfamily
products	array	List of product id of the sub-family

Structure of the array products

Field name	Type	Description
id	string	Product ID
name	string	Product name
price	integer	Price of the product in cents with VAT
tax_rate	string	Product VAT
image_url	string	Image url of the product
is_alcohol	integer	Alcoholic product with value 1, value 0

		non-alcoholic
description	string	Product description
attributes_groups	array	List of attribute groups ids
availability	array	Array of per-day availability. Each item has enabled (boolean), day_of_week (monday to sunday) and time_periods: an array of {start_time, end_time} objects, with times in HH:MM format (e.g. 09:00).
packaging	string	Packaging ID

Structure of the array attributes

Field name	Type	Description
id	string	Attribute ID
name	string	Attribute name
selected_by_default	boolean	Selected by default
price_impact	string	Price impact of the attribute, in decimal euros (e.g. 2.5)
tax_rate	string	Iva of the attribute
attribute_groups	array	Array of attribute groups ids
packaging	string	Packaging ID

Structure of the attribute_groups array

Field name	Type	Description
id	string	Attribute group ID
name	string	Name of the attribute group
min	integer	Minimum number of attribute groups that can be chosen
max	integer	Maximum number of attribute groups that can be chosen

attributes	array	Array of attribute ids
------------	-------	------------------------

Structure of the packagings array

Field name	Type	Description
id	string	Packaging ID
reference_name	string	Name of the packaging
price	integer	Price of packing in cents

Endpoint_menu (defined in the integrator platform for the App)

In the file 'menu_example' is an example of a menu structure for your synchronisation, which will be sent to the menu endpoint.

Example of a successful expected response (statusCode 204)

```
{
  "body": "",
  "header": {
    "statusCode": 204,
    "statusMessage": "No Content",
    "status": "HTTP/1.1 204 No Content"
  }
}
```

Example of response with error (statusCode 400)

```
{
  "body": {
    "error": "Field validation for 'families' failed on the 'required' tag"
  },
  "header": {
    "statusCode": 400,
    "statusMessage": "Bad Request",
  }
}
```

```

    "status": "HTTP/1.1 400 Bad Request"
  }
}

```

Sync Menu via ZSROI

This endpoint allows you to synchronize the menu with ZSROI via an HTTP request. The synchronization process is asynchronous. Once it completes, a final status notification will be sent to BMS.

Please note that there is a daily request limit. If this limit is reached, a specific error message will be returned.

Structure of the sync menu object

Field name	Description
client_id	Client_id available on the portal https://developer.zonesoft.org
store	Store ID

Example request

```

curl --location 'https://zsroi.zonesoft.org/v1.0/integration/syncMenu' \
--header 'Content-Type: application/json' \
--header 'X-Integration-Signature: Signature' \
--header 'Authorization: YOUR_APP_KEY' \
--data '{
  "client_id": "A73FEE38076757E",
  "store": 5
}'

```

Example of a successful response (statusCode 201)

```
{
  "body": {
    "Response": {
      "StatusCode": 201,
      "StatusMessage": "Created",
      "Content": null
    }
  },
  "header": {
    "statusCode": 204,
    "statusMessage": "No Content",
    "status": "HTTP/1.1 204 No Content"
  }
}
```

Example of response with error (statusCode 422)

```
{
  "body": {
    "Response": {
      "StatusCode": 422,
      "StatusMessage": "Unprocessable Entity",
      "Content": [
        {
          "property": "client_id",
          "error": "Max sync per day (5) reached"
        }
      ]
    }
  },
  "header": {
    "statusCode": 422,
    "statusMessage": "Unprocessable Entity",
    "status": "HTTP/1.1 422 Unprocessable Entity"
  }
}
```

Example of response with error (statusCode 400)

```
{
  "body":{
    "Response": {
      "StatusCode": 400,
      "StatusMessage": "Bad Request",
      "Content": null
    }
  }
  "header": {
    "statusCode": 400,
    "statusMessage": "Bad Request",
    "status": "HTTP/1.1 400 Bad Request"
  }
}
```

Example of response with error (statusCode 401)

```
{
  "body":{
    "Response": {
      "StatusCode": 401,
      "StatusMessage": "Unauthorized",
      "Content": null
    }
  }
  "header": {
    "statusCode": 401,
    "statusMessage": "Unauthorized",
    "status": "HTTP/1.1 401 Unauthorized"
  }
}
```

Orders

Orders is the ZSROI functionality that handles orders generated in the integrated App and then routes them to the POS.

Structure of the Orders object

Field name	Type	Description	Required
order_id	string	Order ID	Yes
store_id	string	Your 'client_id' available at https://developer.zonesoft.org	Yes
type_order	string	Type of order -Allowed values: • DELIVERY • PICKUP	Yes
order_time	string	Date and time the order was accepted by the integrating entity in the format: yyyy-MM-dd HH:mm:ss	Yes
estimated_pickup_time	string	Estimated date and time the courier will arrive in the format yyyy-MM-dd HH:mm:ss	Yes
currency	string	Currency code (EUR)	Yes
allergy_info	string	Customer allergy information	No
delivery_fee	integer	Delivery price in cents	No
estimated_total_price	integer	Estimated total price of products and attributes of the order in cents with VAT. The delivery tax is not included	No
courier	object	Object containing information about the courier.	No
customer	object	Object that contains customer information	No
products	array	Array containing the products of the order. One product per object	Yes
obs	string	Observations / Notes	No

orderIsAlreadyPaid	boolean	Information if the order has already been paid for	Yes
payment_type	integer	Payment type Allowed values: 1 - Money 2 - Banker's Cheque 3 - Debit card 4 - Credit card 9 - Multiple Payments (from version 2024.0.1.21 of ZSRest) 28 - Uber Eats 36 - Glovo 39 - Bolt Food 41 - Flipdish	No
delivery_address	object	Delivery address information	Yes
is_picked_up_by_customer	boolean	Boolean to indicate if the order will be picked up by a customer	Yes
discounted_products_total	integer	Total discount of the products - the sum of the 'discount' field of each product which should take into account the field 'attributes[].discount'	Yes
total_customer_to_pay	integer	Total amount to be paid in cents by the customer	Yes
channel_name	string	Name of the origin order supplier. Must use the following: #INTEGRATOR#PROVIDER. Example: #ZONESOFT#UBER_EATS	No
scheduled_order	integer	Unix timestamp for a scheduled order. 0 or absent = immediate order. For scheduled orders the POS availability/ping check is bypassed.	No
loyalty	string	Base64-encoded loyalty/customer code.	No
payment_charges	object	Object with the various charges that make up the payment for an order	No
pick_up_code	object	Pick up code the courier uses to identify order pick up	No

Courier (object)

Field name	Type	Description	Required
name	string	Name of courier	Yes
phone_number	string	Phone number of courier	Yes
license_plate	string	Registration of the courier	Yes

Customer (object)

Field name	Type	Description	Required
name	string	Client's name	Yes
phone_number	string	Customer phone number	No
nif	string	NIF	Yes
email	string	Client's email	No

Produts(array)

Field name	Type	Description	Required
id	string	Product id	Yes
quantity	integer	Order product quantity	Yes
name	string	Product name	Yes
price	integer	Price of the product in cents with VAT	Yes
discount	integer	Product discount in cents	Yes
obs	string	Product observations	Yes
attributes	array	Array containing the product attributes	Yes
packaging	object	Packaging	Yes

Delivery_address(object)

Field name	Type	Description	Required
label	string	Address	Yes
latitude	string	Latitude coordinate	Yes

longitude	string	Longitude coordinate	Yes
street_address	string	Additional street address line	No
obs	string	Delivery observations / notes	No

Attributes(array)

Field name	Type	Description	Required
Id	string	Attribute ID	Yes
quantity	integer	Quantity of the order attribute	Yes
name	string	Attribute name	Yes
price	integer	Price of the attribute in cents with vat	Yes
discount	integer	Product discount in cents	Yes
obs	string	Attribute observations	No
packaging	object	Packaging	Yes

Packaging(object)

Field name	Type	Description	Required
id	string	Packaging ID	Yes
reference_name	string	Name of the packaging	Yes
price	int	Price of packing in cents	Yes

Payment_charges(object)

Field name	Type	Description	Required
total	integer	Total cost of the order, including taxes and fees in cents	Yes
sub_total	integer	Total cost of each product in cents without taxes	Yes
tax	integer	Total cost of fees in cents	Yes
total_fee	integer	Total fees charged on the order that	Yes

		will be paid to the Merchant	
total_fee_tax	integer	The total tax charged on the total rate in cents	Yes
bag_fee	integer	The fee charged to provide a bag(s) for the order in cents	Yes
delivery_fee	integer	The fee charged to deliver the package in cents	Yes
delivery_fee_tax	integer	Tax charged on delivery fee in cents	Yes
small_order_fee	integer	The fee charged for low-value orders in cents	Yes
small_order_fee_tax	integer	Tax charged for low-value parcels in cents	Yes
pick_and_pack_fee	integer	Fee charged for the collection of parcels in cents	Yes
pick_and_pack_fee_tax	integer	Package collection tax in cents	Yes
tip	integer	Total tips in cents	Yes

In the file 'order_example' is an example of an order that the Zone Soft API expects to receive.

Endpoint that should be used for sending orders:

- <https://zsroi.zonesoft.org/v1.0/integration/order>

Note: In the 'Authorization' header you must send your APP-KEY found on the portal:
<https://developer.zonesoft.org/login>

In the 'X-Integration-Signature' header you must send the request signature with your APP-SECRET that can also be found on the portal.

Example of how the request should be signed:

```
hash_hmac('SHA256', $payload, $app_secret);
```

Example of the response of the Zone Soft API on receiving the order if it is saved successfully

```
{
```

```
"Response": {  
  "StatusCode": 201,  
  "StatusMessage": "Created",  
  "Content": null  
}
```

Example of Zone Soft API response when the order is received and there is no menu on the Zone Soft side, with the same ids as the order.

```
{  
  "Response": {  
    "StatusCode": 422,  
    "StatusMessage": "Unprocessable Entity",  
    "Content": "order_rejected_pos_menu_not_found"  
  }  
}
```

Example of response from the Zone Soft API when order is received and the POS is not connected

```
{  
  "Response": {  
    "StatusCode": 422,  
    "StatusMessage": "Unprocessable Entity",  
    "Content": "order_rejected_pos_ping_failed"  
  }  
}
```

Example of the response from the Zone Soft API when the order is received and the integration is not active

```
{  
  "Response": {  
    "StatusCode": 422,  
    "StatusMessage": "Unprocessable Entity",  
    "Content": "order_rejected_pos_missing_integration_service"  
  }  
}
```

Example of response from the Zone Soft API when one or more products are not found in the Zone Soft menu

```
{
  "Response": {
    "StatusCode": 422,
    "StatusMessage": "Unprocessable Entity",
    "Content": "order_rejected_unknown_items: 1978,1979"
  }
}
```

Example of response from the Zone Soft API when the order fields fail validation

```
{
  "Response": {
    "StatusCode": 422,
    "StatusMessage": "Unprocessable Entity",
    "Content": "order_rejected_bad_structure"
  }
}
```

Note: the 400 Bad Request (Content: null) below is returned when core fields are missing from the request: customer, order_id, products, store_id, courier, type_order.

Example of response from the Zone Soft API when the order is received and the structure is not as expected

```
{
  "Response": {
    "StatusCode": 400,
    "StatusMessage": "Bad Request",
    "Content": null
  }
}
```

Endpoint_order_status (defined in the integrator platform for the App)

ZSROI will send status updates of orders received to this endpoint: 1 request per order.

Supported states: 'accept', 'decline', 'ready', 'delivered', 'remind'.

```
curl --location --request POST 'https://example.com/order/status \
--header 'Content-Type: application/json' \
--header 'Authorization: TOKEN' \
--data-raw '{
  "order_id": "123123",
  "status": "ready"
}'
```

When the status is 'accept', an additional field named "estimated_time" (time in second) is included. This field is optional.

```
curl --location --request POST 'https://example.com/order/status \
--header 'Content-Type: application/json' \
--header 'Authorization: TOKEN' \
--data-raw '{
  "order_id": "123123",
  "status": "accept",
  "estimated_time": 600
}'
```

Example of a successful response (statusCode 204)

```
{
  "body": "",
  "header": {
    "statusCode": 204,
    "statusMessage": "No Content",
    "status": "HTTP/1.1 204 No Content"
  }
}
```

Example of response with error (statusCode 404)

```
{
  "body": {
    "code": "resource_not_found",
    "message": "Order not found"
  },
  "header": {
    "statusCode": 404,
    "statusMessage": "Not Found",
    "status": "HTTP/1.1 404 Not Found"
  }
}
```


Endpoint_pos_status

The POS endpoints have the functionality to get information from the POS, to put the POS online or offline.

Note: The suffix 'closing' is added to endpoint_pos_status for PUT and DELETE requests.

POS data - Get POS status

```
curl --location --request GET 'https://example.com/pos/status' \
--header 'Content-Type: application/json' \
--header 'Authorization: TOKEN' \
--data-raw ""
```

Example of a successful response (statusCode 200)

```
{
  "body": "",
  "header": {
    "statusCode": 200,
    "statusMessage": "OK",
    "status": "HTTP/1.1 200 OK"
  }
}
```

POS online - Putting the POS online to receive orders

```
curl --location --request DELETE 'https://example.com/pos/status/closing' \
--header 'Content-Type: application/json' \
--header 'Authorization: TOKEN' \
--data-raw ""
```

Example of a successful response (statusCode 204)

```
{
  "body": "",
  "header": {
    "statusCode": 204,
    "statusMessage": "No Content",
    "status": "HTTP/1.1 204 No Content"
  }
}
```

POS offline - Put the POS offline to not receive orders.

```
curl --location --request PUT 'https://example.com/pos/status/closing' \
--header 'Content-Type: application/json' \
--header 'Authorization: TOKEN' \
--data-raw ""
```

Example of a successful response (statusCode 204)

```
{
  "body": "",
  "header": {
    "statusCode": 204,
    "statusMessage": "No Content",
    "status": "HTTP/1.1 204 No Content"
  }
}
```

Table Reservation

To make a table reservation, you must follow the authentication process detailed in the "Login" section. Further details on reservations are available in the linked document:

https://cdn1.zonesoft.org/docs/zsroi/v1/TablesReservations_API.pdf

Get Tables

Functionality to get the information from the tables to the serial and store

exemplo:

```
curl --location 'https://zsroi.zonesoft.org/v1.0/integration/tables/base_encode(seria+loja)' \
--header 'X-Integration-Signature: Signature' \
--header 'Authorization: YOUR_APP_KEY' \
--data "
```